Planning for a measles outbreak at UVA

Why are we doing this? The purpose of this case study is to connect the material that we've learned about network structure to a real-world system. This is meant to be open-ended and harness your problem-solving skills but from a network science perspective.

How will you be assessed? Half of the credit will be based on the pre-class assignment and half of the credit will be based on your team's post-class written answers. The purpose of the pre-class assignment will be to come to class prepared, so for this reason, **no late pre-class assignments will be accepted**. I will be grading the assignments based on **the evidence used to justify your answer**. It is most important that you explain *why* your proposed solution/answer makes sense.

What will you submit?

- **Before class on November 18th ("Pre-class assignment"):** Submit your responses to the pre-class assignment as a PDF to Canvas with (1) answers to each of the questions and (2) a figure generated from running the analysis notebook provided to you.
- **Post-class ("Group assignment"):** Submit your group discussion as a PDF to Canvas. Each group can submit a single PDF.

Format requirements

Pre-class assignment

- The title of the case study
- Your name
- The answers to the three questions, clearly delineated with headers (the homework template would be helpful for this), and the generated figure of the SIR model from the notebook.

Group assignment

- The title of the case study
- The members of your team
- The answers to the three questions, clearly delineated with headers (the homework template would be helpful for this).

The assignment

Before class on November 18:

- Watch the "Measles: Understanding the most contagious preventable disease" YouTube video and answer the following:
  - What are common symptoms of the measles?
  - Why is measles so transmissible?
- Read the Center for Disease Prevention and Control (CDC) page titled "Measles Cases and Outbreaks" and answer the following:
  - What is the range in MMR vaccination rates for kindergarten-aged children in the US?
  - How many measles cases occurred in the US in 2025?
- Read the Wikipedia page titled "Herd Immunity#Theoretical basis" and answer the following:

  - What should the vaccination rate be for herd immunity against measles ($R_0 \approx$ 12-18)?
- Read the CDC page titled "Behind the Model: How Disease Modeling Supported Decision-Making in a Local Measles Outbreak Response" and answer the following:
  - How was a mathematical model used to inform public health officials?
- Download the `uva_contact_network` dataset and the Jupyter Notebook from the Case Studies page and save the first figure in the notebook. Note: you'll need to pip install EoN for the notebook to work.

During the November 18 class:

Discuss the following questions in your assigned groups:

- Your answers to the pre-class assignment.
- Using the notebook provided to you:
  - Based on the nationwide vaccination rates, how large would you expect an outbreak at UVA as a best-case and worst-case scenario?
  - What should vaccinations rates be to prevent large-scale transmission? How does this match your results from the pre-class assignment?
  - What intervention strategy would you recommend for UVA if there is a measles outbreak at UVA (online classes vs. masking)? What are the pros and cons of each?

After the November 18 class:

Write up your discussion as a 1-page document (single-spaced, 1 inch margins, 11 point font) as a briefing to Pres. Mahoney on what to expect for a measles outbreak at UVA and your recommended intervention strategy. Your write-up must incorporate your group's answers to the in-class questions.